Building and evaluation of a PBPK model for Risperidone in adults

Version	**evaluation-OSP12.2**
based on *Model Snapshot* and *Evaluation Plan*	https://github.com/Open-Systems-Pharmacology/Risperidone-Model/releases/tag/vevaluation
OSP Version	12.2
Qualification Framework Version	3.5

This evaluation report and the corresponding PK-Sim project file are filed at:

https://github.com/Open-Systems-Pharmacology/OSP-PBPK-Model-Library/

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1 Introduction

Risperidone is an atypical antipsychotic used for schizophrenia and other psychiatric indications. It is administered orally and is converted to the active metabolite 9-hydroxyrisperidone, also known as paliperidone. CYP2D6 is an important metabolic pathway for risperidone, while CYP3A4 and transporter processes contribute to parent and metabolite disposition.

This risperidone model is intended to describe plasma concentration-time profiles of risperidone, 9-hydroxyrisperidone, and active moiety across CYP2D6 phenotype and activity-score groups. It supports evaluation of CYP2D6 drug-gene interactions and interacting-drug scenarios where risperidone or 9-hydroxyrisperidone exposure is clinically relevant.

The risperidone parent-metabolite PBPK model was originally developed by Kneller 2020. The model was subsequently used in the CYP2D6 activity-score framework by Rüdesheim 2022 and in the CYP2D6 drug-drug-gene interaction network by Rüdesheim 2025. The clinical data include oral single-dose and multiple-dose administration of risperidone, CYP2D6 phenotype or activity-score stratified groups, and studies with CYP3A or P-gp modulators.

The presented model includes the following features:

- risperidone and 9-hydroxyrisperidone as parent and metabolite compounds,
- CYP2D6-mediated risperidone metabolism with activity-score-dependent k_{cat} values,
- CYP3A4-mediated metabolism and residual sink pathways,
- P-gp transport for risperidone and 9-hydroxyrisperidone,
- renal filtration for parent and metabolite,
- oral tablet administration in the evaluated clinical studies.

2 Methods

2.1 Modeling strategy

The general concept of building a PBPK model has previously been described by Kuepfer et al. (Kuepfer 2016). Relevant information on anthropometric and physiological parameters in adults was gathered from the literature and incorporated into PK-Sim as default values for adult simulations (Willmann 2007).

The applied activity and variability of plasma proteins and active processes integrated into PK-Sim are described in the publicly available PK-Sim Ontogeny Database or otherwise referenced for the specific process.

The risperidone model was developed as a parent-metabolite model for risperidone and 9-hydroxyrisperidone. The original model by Kneller 2020 described oral risperidone pharmacokinetics by CYP2D6 phenotype. Rüdesheim 2022 then used the model in a CYP2D6 activity-score framework, where CYP2D6 metabolic capacity is represented by activity-score-dependent k_{cat} values.

Clinical studies used for model building covered oral risperidone administration and included data for risperidone and 9-hydroxyrisperidone where available. Model verification used independent oral study arms, including CYP2D6 poor-, normal-, and higher-activity groups and interacting-drug scenarios. Because the active moiety is the sum of parent and metabolite exposure, model performance has to be interpreted separately for risperidone, 9-hydroxyrisperidone, and active moiety.

The model-building workflow first established the parent disposition and the formation of 9-hydroxyrisperidone. CYP2D6-related parameters were then used to describe the shift from parent exposure toward metabolite exposure across phenotype and activity-score groups. Verification focused on whether the final parameter set described independent study arms without additional study-specific adjustment.

The model contains CYP2D6 and CYP3A4 metabolism, P-gp transport, renal filtration, and residual clearance processes. CYP2D6 poor-metabolizer activity is represented by setting CYP2D6 k_{cat} to zero, while non-zero activity-score groups use fitted or scaled CYP2D6 k_{cat} values.

The evaluated applications are oral risperidone applications with risperidone, 9-hydroxyrisperidone, and active-moiety observations. The major proteins and processes represented explicitly are CYP2D6, CYP3A4, P-gp, plasma protein binding, passive renal filtration, and unspecific hepatic clearance for the metabolite. The parent-metabolite structure is central to the evaluation because a model can describe active moiety while still showing bias in the parent or metabolite alone.

The report therefore presents model diagnostics for the parent and metabolite concentrations and links the clinical-data table to model-building and verification profiles. The intended interpretation is the plausibility of risperidone and 9-hydroxyrisperidone exposure across CYP2D6 status, not a separate reoptimization for each clinical study.

Details about input data are provided in Section 2.2. Details about the structural model and assumptions are provided in Section 2.3.

2.2 Data used

2.2.1 In vitro and physicochemical data

The table below summarizes the drug-dependent inputs documented for the risperidone model. Parameter values are listed for the parent compound and 9-hydroxyrisperidone where they are relevant for interpreting the parent-metabolite simulations.

Parameter	Unit	Value	Source	Description
Risperidone MW	g/mol	410.48	Kneller 2020	Parent compound size used in concentration conversions.
Risperidone pK_a	-	8.76, 3.11	Kneller 2020	Basic and acidic ionization constants.
Risperidone f_u	%	17.50	Kneller 2020	Plasma binding input for parent drug.
Risperidone CYP2D6 K_m to 9-hydroxyrisperidone	µmol/L	1.10	Kneller 2020	CYP2D6 affinity parameter for metabolite formation.
Risperidone CYP2D6 k_{cat} EM to 9-hydroxyrisperidone	1/min	1.07	Optimized	CYP2D6 metabolic capacity in extensive metabolizers.
Risperidone P-gp K_m	µmol/L	26.30	Kneller 2020	P-gp affinity parameter.
Risperidone P-gp k_{cat}	1/min	12.72	Optimized	P-gp transport capacity.
9-Hydroxyrisperidone f_u	%	29.00	Kneller 2020	Plasma binding input for the active metabolite.
9-Hydroxyrisperidone P-gp K_m	µmol/L	149.60	Rüdesheim 2022	P-gp affinity parameter for the active metabolite.
9-Hydroxyrisperidone unspecific CL_{hep}	1/min	0.08	Optimized	Residual metabolite elimination.
GFR fraction	-	1.00	Assumed	Passive glomerular filtration fraction for parent and metabolite.

The 9-hydroxyrisperidone P-gp K_m reported in the publication and supplement by Rüdesheim 2022 is erroneous. The model retains the correct value of 149.6 µmol/L.

2.2.2 Clinical data

The evaluation uses 19 observed-data records for risperidone and 9-hydroxyrisperidone in peripheral venous blood plasma. The evaluation plan assigns 6 simulations to model building and 6 simulations to model verification.

Model-building clinical data:

Publication	Arm / Treatment / Information used for model building
Novalbos 2010	Plasma PK profiles in adults after oral, single-dose administration of 1 mg risperidone with CYP2D6 AS = 2 status.
Bondolfi 2002	Plasma PK profiles in adults after oral, multiple dose administration of 2 mg risperidone with CYP2D6 poor- and extensive-metabolizer status.
Markowitz 2002	Plasma PK profiles in adults after oral, single-dose administration of 1 mg risperidone with CYP2D6 extensive-metabolizer status.
Darwish 2015	Plasma PK profiles in adults after oral, single-dose administration of 2 mg risperidone with CYP2D6 extensive-metabolizer status.
Mahatthanatrakul 2012	Plasma PK profiles in adults after oral, single-dose administration of 2 mg risperidone with risperidone and 9-hydroxyrisperidone measurements.

Model-verification clinical data:

Publication	Arm / Treatment / Information used for model verification
Novalbos 2010	Plasma PK profiles in adults after oral, single-dose administration of 1 mg risperidone with CYP2D6 AS = 0, AS = 1, and AS = 3 status.
Nakagami 2005	Plasma PK profiles in adults after oral, single-dose administration of 1 mg risperidone with risperidone and 9-hydroxyrisperidone measurements.
Mahatthanatrakul 2007	Plasma PK profiles in adults after oral, single-dose administration of 4 mg risperidone.
Kim 2008	Plasma PK profiles in adults after oral risperidone administration with rifampin pretreatment.

2.3 Model parameters and assumptions

2.3.1 Absorption

The model includes oral administration of risperidone. Oral absorption is represented with compound-specific intestinal permeability and formulation settings. The risperidone intestinal permeability was optimized, while the 9-hydroxyrisperidone intestinal permeability was calculated.

The clinical studies used for model building and verification are oral tablet studies, so the evaluation focuses on oral risperidone absorption and subsequent parent-metabolite disposition.

No intravenous risperidone data are included in the evaluation, so oral absorption and first-pass metabolism are coupled in the clinical concentration-time profiles. The `Specific intestinal` `permeability` parameter is therefore interpreted together with the metabolic parameters that determine parent and metabolite exposure after oral dosing.

The same oral absorption assumptions are used across CYP2D6 status groups. Phenotype and activity-score differences are assigned to metabolic capacity rather than to study-specific changes in absorption.

2.3.2 Distribution

Risperidone and 9-hydroxyrisperidone are represented with compound-specific plasma protein binding. Fraction unbound values of 17.50% for risperidone and 29.00% for 9-hydroxyrisperidone were used as summarized in Section 2.2.1.

Partition coefficients were calculated with the Rodgers and Rowland method. The parent-metabolite structure means that distribution affects both the parent concentration-time profile and the active-moiety profile.

The model uses separate physicochemical and distribution properties for risperidone and 9-hydroxyrisperidone. This is important because active moiety is a composite endpoint, while tissue distribution and plasma binding are compound-specific. The active-moiety comparison should therefore be read together with the individual parent and metabolite profiles.

Distribution was not fitted separately by phenotype. CYP2D6 status affects the rate of risperidone conversion to 9-hydroxyrisperidone, while the same adult physiology and compound-specific binding assumptions are retained across simulations.

2.3.3 Metabolism and Elimination

Two CYP-mediated pathways, transporter processes, renal filtration, and residual clearance pathways are represented in the model.

- CYP2D6

CYP2D6 converts risperidone to 9-hydroxyrisperidone. CYP2D6 activity is activity-score-dependent. Poor-metabolizer activity is set to zero, while non-zero activity-score groups use fitted or scaled k_{cat} values based on the CYP2D6 activity-score framework.

The CYP2D6 pathway controls both parent depletion and metabolite formation. For this reason, CYP2D6 parameterization affects risperidone, 9-hydroxyrisperidone, and active-moiety profiles in different directions. Parent-only diagnostics are not sufficient to evaluate this model.

- CYP3A4 and sink pathways

CYP3A4 contributes to risperidone metabolism. Additional sink pathways are included to capture clearance not explicitly assigned to CYP2D6-mediated 9-hydroxyrisperidone formation.

These pathways provide CYP2D6-independent clearance and prevent poor-metabolizer simulations from depending solely on renal elimination. Their interpretation is structural and empirical, because they represent clearance components not resolved as individual measured metabolites in the clinical evaluation.

- P-gp and renal elimination

P-gp transport is implemented for risperidone and 9-hydroxyrisperidone. Both compounds include passive renal filtration with a `GFR fraction` of 1. The metabolite also includes unspecific hepatic clearance as summarized in Section 2.2.1.

P-gp transport was retained for both parent and metabolite because it can affect intestinal and renal handling. Passive renal filtration uses the compound-specific fraction unbound and adult renal physiology. The additional metabolite clearance pathway captures elimination of 9-hydroxyrisperidone not represented by filtration alone.

3 Results and Discussion

The PBPK model for risperidone was developed and evaluated with clinical pharmacokinetic data after oral administration. The evaluation covers single-dose and multiple-dose oral administration, risperidone and 9-hydroxyrisperidone plasma concentration-time profiles, CYP2D6 poor-, extensive-, and activity-score stratified groups, and interaction settings relevant to CYP3A and P-gp.

The model-building data supported the parent-metabolite structure, CYP2D6-mediated 9-hydroxylation, CYP3A4 metabolism, P-gp transport, renal filtration, and residual metabolite clearance. Model-building studies included CYP2D6 activity-score or phenotype data from Novalbos 2010, Bondolfi 2002, and Markowitz 2002, as well as studies with parent and metabolite measurements from Darwish 2015 and Mahatthanatrakul 2012. Verification included independent activity-score and interaction studies from Novalbos 2010, Nakagami 2005, Mahatthanatrakul 2007, and Kim 2008.

The model quantifies CYP2D6-mediated formation of 9-hydroxyrisperidone, CYP3A4-mediated metabolism, P-gp transport, passive renal filtration, and residual clearance of the active metabolite. The interpretation of model performance requires separate consideration of parent, metabolite, and active-moiety behavior. Parent risperidone profiles mainly test absorption, P-gp transport, and formation clearance. 9-hydroxyrisperidone profiles additionally test metabolite distribution, renal filtration, and residual hepatic clearance.

The next sections show:

1. the final model input parameters for the building blocks: Section 3.1.
2. the overall goodness of fit: Section 3.2.
3. simulated vs. observed concentration-time profiles for the clinical studies used for model building and for model verification: Section 3.3.

The merged GOF diagnostic over the included concentration observations gives GMFE values of 1.30 for risperidone and 1.31 for 9-hydroxyrisperidone. The diagnostic output does not provide a separately calculated overall GMFE. These values show close population-level agreement. Parent and metabolite performance should nevertheless be considered separately when interpreting active-moiety predictions, because compensation between risperidone and 9-hydroxyrisperidone can mask analyte-specific bias.

Rüdesheim 2022 reported strong DGI ratio performance for risperidone, with 7 of 7 AUClast ratios and 7 of 7 Cmax ratios within the prediction success limits. The reported GMFE values were 1.11 for AUClast ratios and 1.16 for Cmax ratios. These DGI ratio metrics support the activity-score implementation, while the concentration-time diagnostics below remain necessary for evaluating the full parent-metabolite model.

The concentration-time profiles should be reviewed by CYP2D6 activity group and analyte. CYP2D6 poor-metabolizer profiles test whether CYP3A4 metabolism and residual processes can describe parent clearance when CYP2D6 activity is absent. Higher activity-score groups test whether increased CYP2D6 k_{cat} values improve parent-to-metabolite conversion without overpredicting 9-hydroxyrisperidone. Interaction studies involving rifampin or other modulators additionally test whether CYP3A and P-gp assumptions remain plausible beyond the base oral dosing studies.

The model is adequate for adult oral risperidone simulations within the represented doses, CYP2D6 groups, and measured analytes. Remaining interpretation should be cautious for unrepresented formulations, pediatric populations, and scenarios where the active moiety is used without checking parent and metabolite profiles separately.

3.1 Risperidone final input parameters

The compound parameter values of the final PBPK model are illustrated below.

Compound: Risperidone

Parameters

Name	Value	Value Origin	Alternative	Default
Solubility at reference pH	0.171 mg/ml	Publication-Other-Kneller et al. 2020	Measurement	True
Reference pH	7.3	Publication-Other-Kneller et al. 2020	Measurement	True
Lipophilicity	2.4 Log Units	Publication-Other-Kneller et al. 2020	LogP	True
Fraction unbound (plasma, reference value)	17.5 %	Publication-Other-Kneller et al. 2020	Measurement	True
Specific intestinal permeability (transcellular)	8.0437234696E-06 cm/min	Parameter Identification-Parameter Identification-Optimized	Optimized	True
F	1			
Is small molecule	Yes			
Molecular weight	410.48 g/mol			
Plasma protein binding partner	α1-acid glycoprotein			

Calculation methods

Name	Value
Partition coefficients	Rodgers and Rowland
Cellular permeabilities	PK-Sim Standard

Processes

Metabolizing Enzyme: CYP2D6-AS=2 (9-hydroxylation)

Molecule: CYP2D6

Metabolite: 9-Hydroxyrisperidone

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0 pmol/min/mg mic. protein	
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	1.1 µmol/l	Other-Manual Fit
kcat	3.1902307094 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-AS=2 (sink)

Molecule: CYP2D6

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0 pmol/min/mg mic. protein	
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	1.1 µmol/l	Other-Manual Fit
kcat	1.9418795623 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP3A4-Okubo 2016 (9-hydroxylation)

Molecule: CYP3A4

Metabolite: 9-Hydroxyrisperidone

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0 pmol/min/mg mic. protein	
Km	61 µmol/l	Publication-Other-Kneller et al. 2020
kcat	0.7 1/min	Publication-Other-Kneller et al. 2020

Metabolizing Enzyme: CYP3A4-Okubo 2016 (sink)

Molecule: CYP3A4

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0 pmol/min/mg mic. protein	
Km	61 µmol/l	Publication-Other-Kneller et al. 2020
kcat	0.15 1/min	Publication-Other-Kneller et al. 2020

Transport Protein: ABCB1-Ejsing 2005

Molecule: ABCB1

Parameters

Name	Value	Value Origin
Transporter concentration	1 µmol/l	
Vmax	0 µmol/l/min	
Km	26.3 µmol/l	
kcat	12.715199036 1/min	Parameter Identification-Parameter Identification-Optimized

Systemic Process: Glomerular Filtration-Assumed

Species: Human

Parameters

Name	Value	Value Origin
GFR fraction	1	

Metabolizing Enzyme: CYP2D6-AS=0 (9-hydroxylation)

Molecule: CYP2D6

Metabolite: 9-Hydroxyrisperidone

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0 pmol/min/mg mic. protein	
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	1.1 µmol/l	

Metabolizing Enzyme: CYP2D6-AS=0 (sink)

Molecule: CYP2D6

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0 pmol/min/mg mic. protein	
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	1.1 µmol/l	

Metabolizing Enzyme: CYP2D6-AS=1 (9-hydroxylation)

Molecule: CYP2D6

Metabolite: 9-Hydroxyrisperidone

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0 pmol/min/mg mic. protein	
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	1.1 µmol/l	Other-Manual Fit
kcat	1.2197940948 1/min	Other-Manual Fit

Metabolizing Enzyme: CYP2D6-AS=1 (sink)

Molecule: CYP2D6

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0 pmol/min/mg mic. protein	
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	1.1 µmol/l	Other-Manual Fit
kcat	0.742483362 1/min	Other-Manual Fit

Metabolizing Enzyme: CYP2D6-AS=3 (9-hydroxylation)

Molecule: CYP2D6

Metabolite: 9-Hydroxyrisperidone

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0 pmol/min/mg mic. protein	
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	1.1 µmol/l	Other-Manual Fit
kcat	5.911309844 1/min	Other-Manual Fit

Metabolizing Enzyme: CYP2D6-AS=3 (sink)

Molecule: CYP2D6

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0 pmol/min/mg mic. protein	
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	1.1 µmol/l	Other-Manual Fit
kcat	3.5981886007 1/min	Other-Manual Fit

Metabolizing Enzyme: CYP2D6-Okubo 2016 (9-hydroxylation)

Molecule: CYP2D6

Metabolite: 9-Hydroxyrisperidone

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0 pmol/min/mg mic. protein	
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	1.1 µmol/l	
kcat	1.07 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-Okubo 2016 (sink)

Molecule: CYP2D6

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0 pmol/min/mg mic. protein	
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	1.1 µmol/l	
kcat	0.67 1/min	Parameter Identification-Parameter Identification-Optimized

Metabolizing Enzyme: CYP2D6-AS=1.25 (9-hydroxylation)

Molecule: CYP2D6

Metabolite: 9-Hydroxyrisperidone

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0 pmol/min/mg mic. protein	
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	1.1 µmol/l	
kcat	1.66 1/min	Other-Manual Fit

Metabolizing Enzyme: CYP2D6-AS=1.25 (sink)

Molecule: CYP2D6

Parameters

Name	Value	Value Origin
In vitro Vmax for liver microsomes	0 pmol/min/mg mic. protein	
Content of CYP proteins in liver microsomes	10 pmol/mg mic. protein	Unknown
Km	1.1 µmol/l	
kcat	1.01 1/min	Other-Manual Fit

Compound: 9-Hydroxyrisperidone

Parameters

Name	Value	Value Origin	Alternative	Default
Solubility at reference pH	0.171 mg/ml	Publication-Other-Kneller et al. 2020	Measurement	True
Reference pH	7.3	Publication-Other-Kneller et al. 2020	Measurement	True
Lipophilicity	2.1 Log Units	Publication-Other-Kneller et al. 2020	LogP	True
Fraction unbound (plasma, reference value)	29 %	Publication-Other-Kneller et al. 2020	Measurement	True
F	1			
Is small molecule	Yes			
Molecular weight	426.48 g/mol			
Plasma protein binding partner	α1-acid glycoprotein			

Calculation methods

Name	Value
Partition coefficients	Rodgers and Rowland
Cellular permeabilities	PK-Sim Standard

Processes

Systemic Process: Total Hepatic Clearance-Assumption

Species: Human

Parameters

Name	Value	Value Origin
Fraction unbound (experiment)	0.26	
Lipophilicity (experiment)	1.76 Log Units	
Plasma clearance	0 ml/min/kg	
Specific clearance	0.0821152637 1/min	Parameter Identification-Parameter Identification-Optimized

Systemic Process: Glomerular Filtration-Assumption

Species: Human

Parameters

Name	Value	Value Origin
GFR fraction	1	

Transport Protein: ABCB1-Ejsing 2005

Molecule: ABCB1

Parameters

Name	Value	Value Origin
Transporter concentration	1 µmol/l	
Vmax	0 µmol/l/min	
Km	149.6 µmol/l	
kcat	0.0057047043998 1/min	Parameter Identification-Parameter Identification-Optimized

3.2 Diagnostic plots

Below you find the goodness-of-fit visual diagnostic plots for PBPK model performance of all data used, as presented in Section 2.2.2 (Clinical data).

The first plot shows observed versus simulated plasma concentration, the second weighted residuals versus time.

3.2.1 Risperidone goodness-of-fit diagnostics

Below you find the goodness-of-fit visual diagnostic plots for risperidone plasma concentration data used in the model evaluation.

The first plot shows observed versus simulated plasma concentration, the second weighted residuals versus time.

Table 3-1: GMFE for Risperidone observed versus simulated plasma concentration-time data

Group	GMFE
Risperidone	1.30

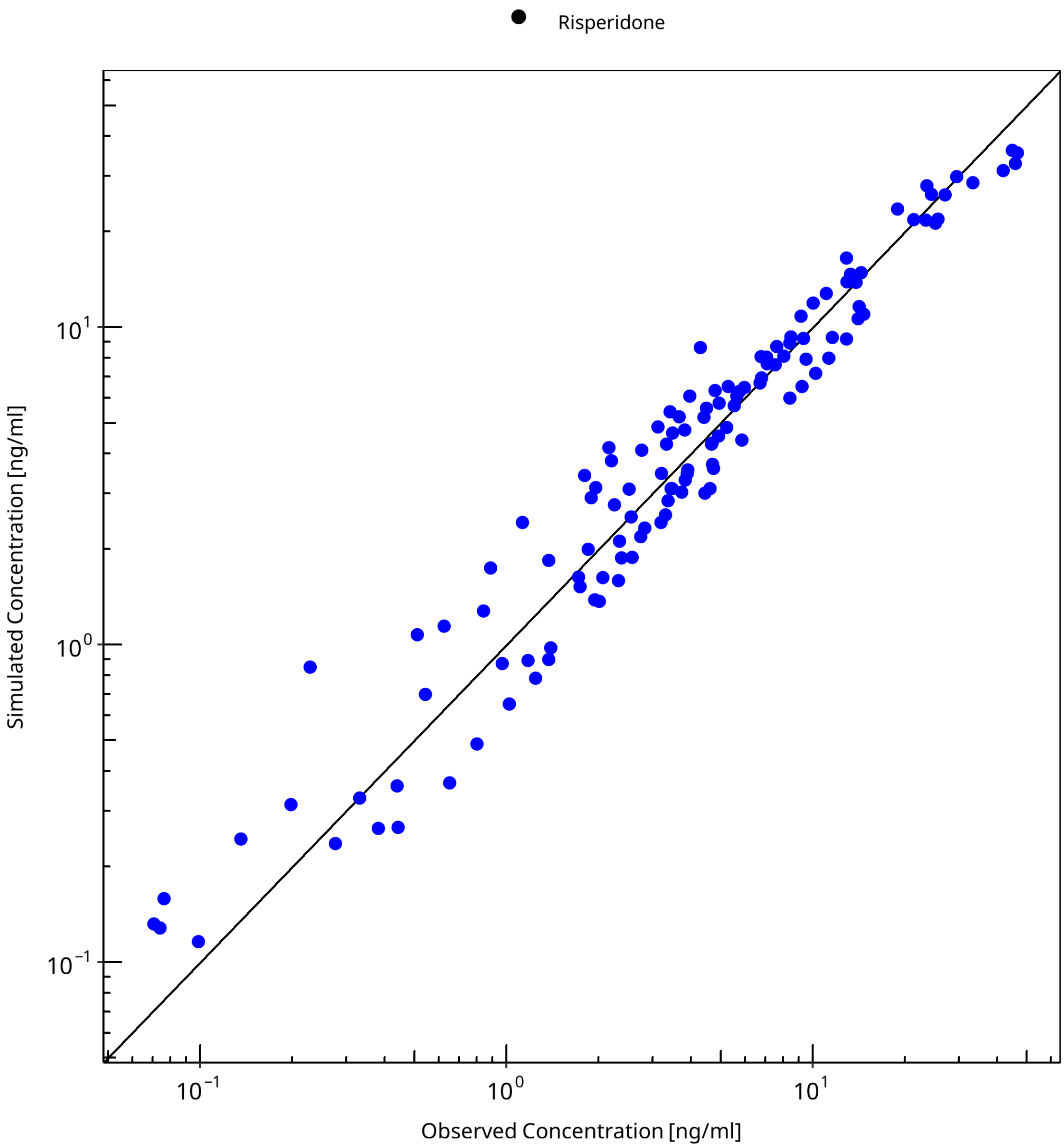

Figure 3-1: Risperidone observed versus simulated plasma concentration-time data

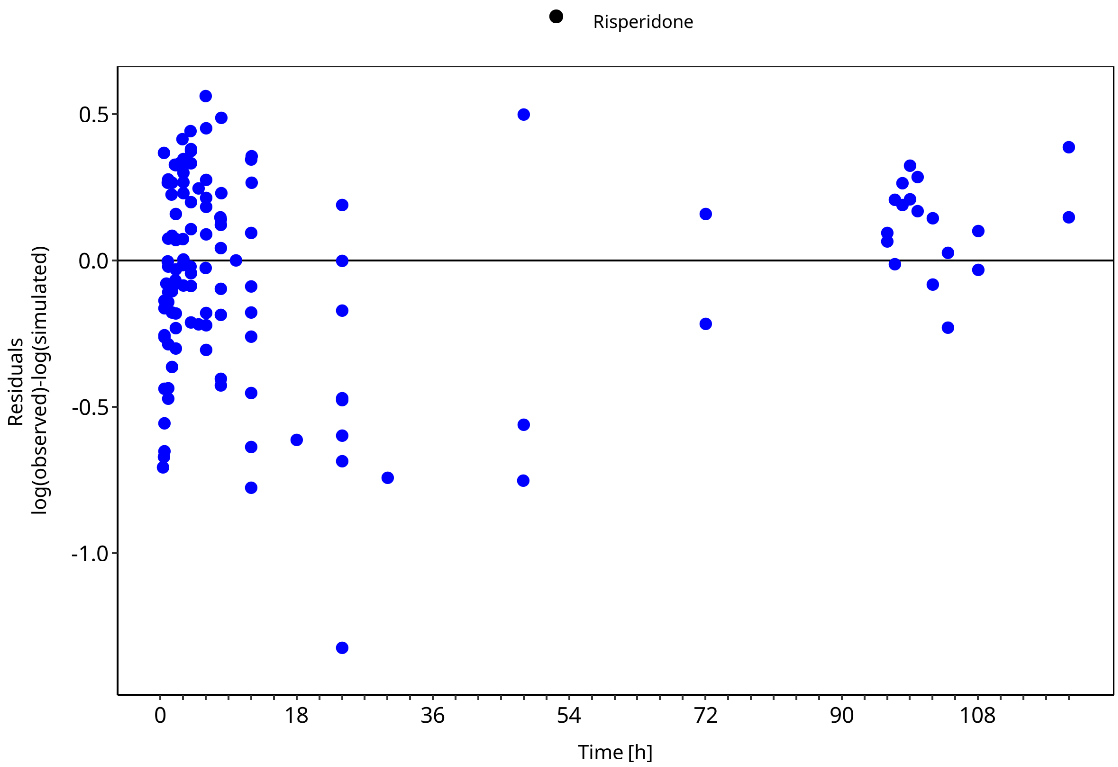

Figure 3-2: Risperidone observed versus simulated plasma concentration-time data

3.2.2 9-Hydroxyrisperidone goodness-of-fit diagnostics

Below you find the goodness-of-fit visual diagnostic plots for 9-hydroxyrisperidone plasma concentration data used in the model evaluation.

The first plot shows observed versus simulated plasma concentration, the second weighted residuals versus time.

Table 3-2: GMFE for 9-Hydroxyrisperidone observed versus simulated plasma concentration-time data

Group	GMFE
9-Hydroxyrisperidone	1.31

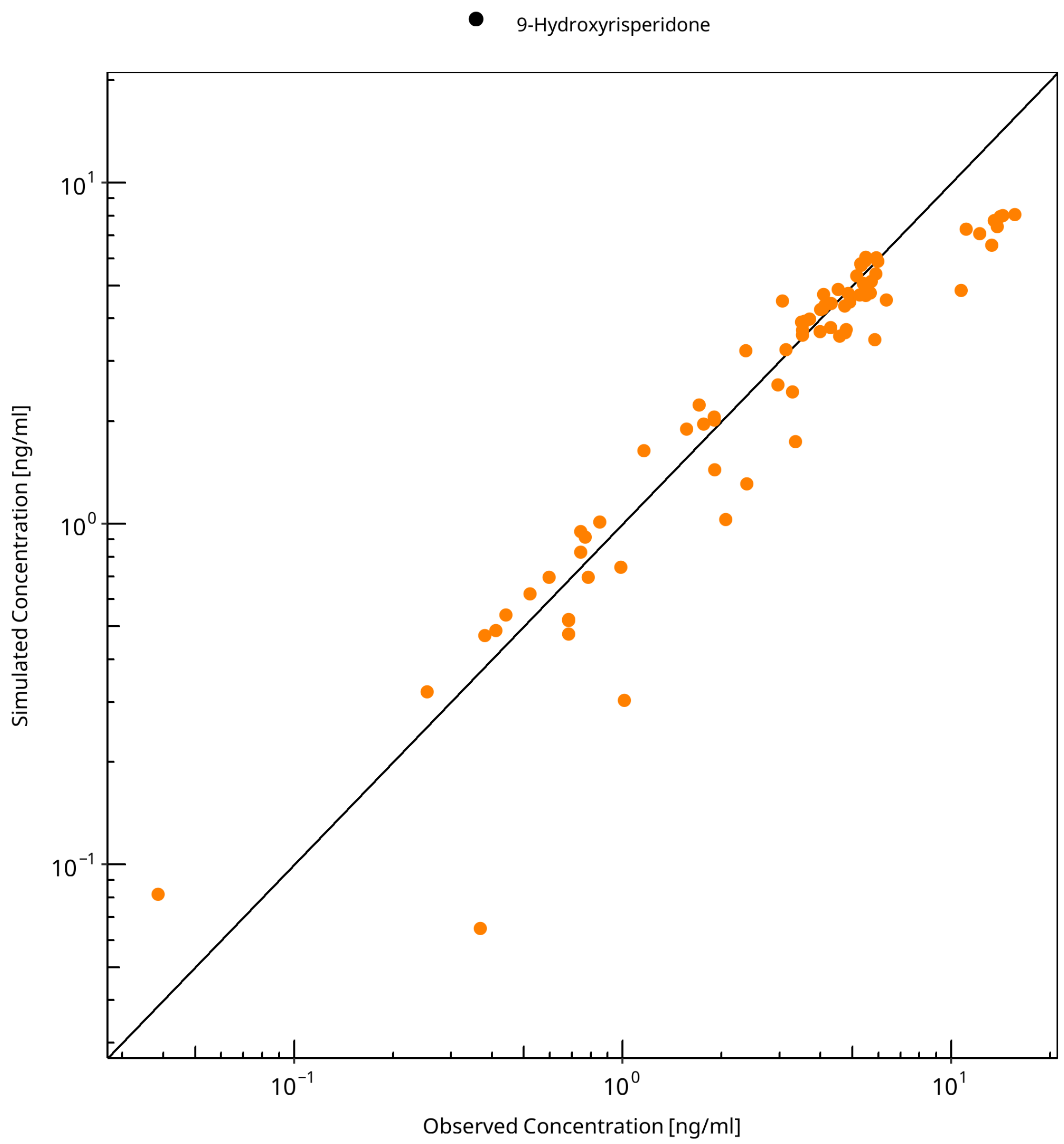

Figure 3-3: 9-Hydroxyrisperidone observed versus simulated plasma concentration-time data

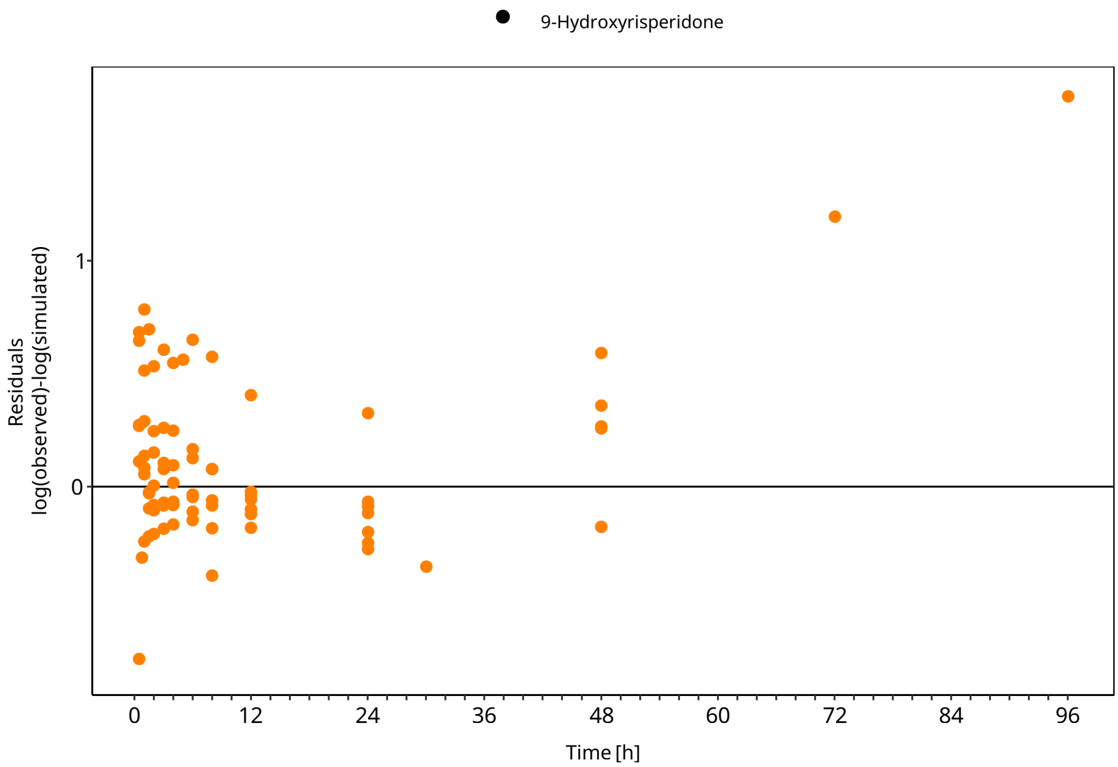

Figure 3-4: 9-Hydroxyrisperidone observed versus simulated plasma concentration-time data

3.3 Concentration-time profiles

Simulated versus observed concentration-time profiles of all data listed in Section 2.2.2 (Clinical data) are presented below.

3.3.1 Model Building

The following model-building profiles show risperidone and 9-hydroxyrisperidone data used to establish parent disposition and CYP2D6-mediated metabolite formation. Parent and metabolite observations are interpreted separately.

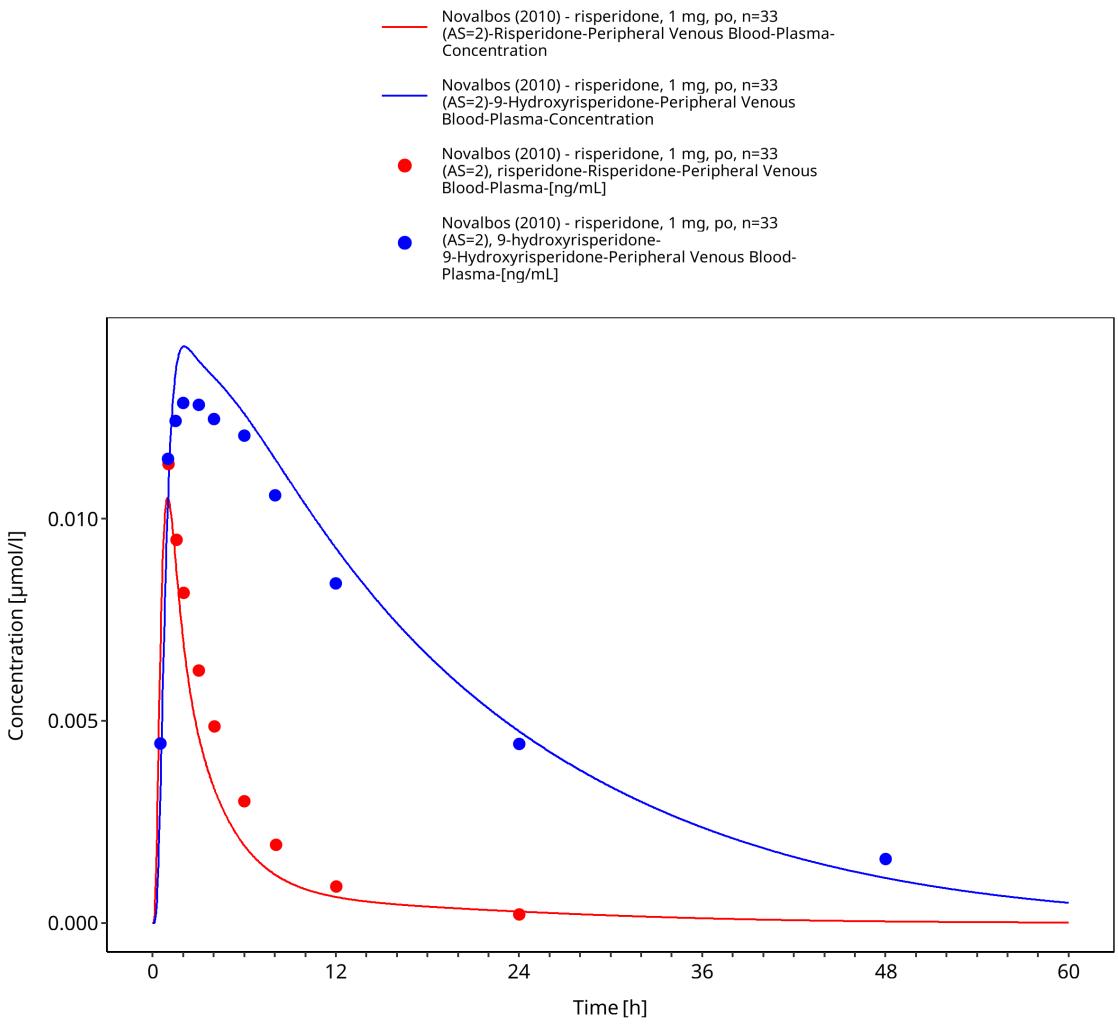

Figure 3-5: Time Profile Analysis

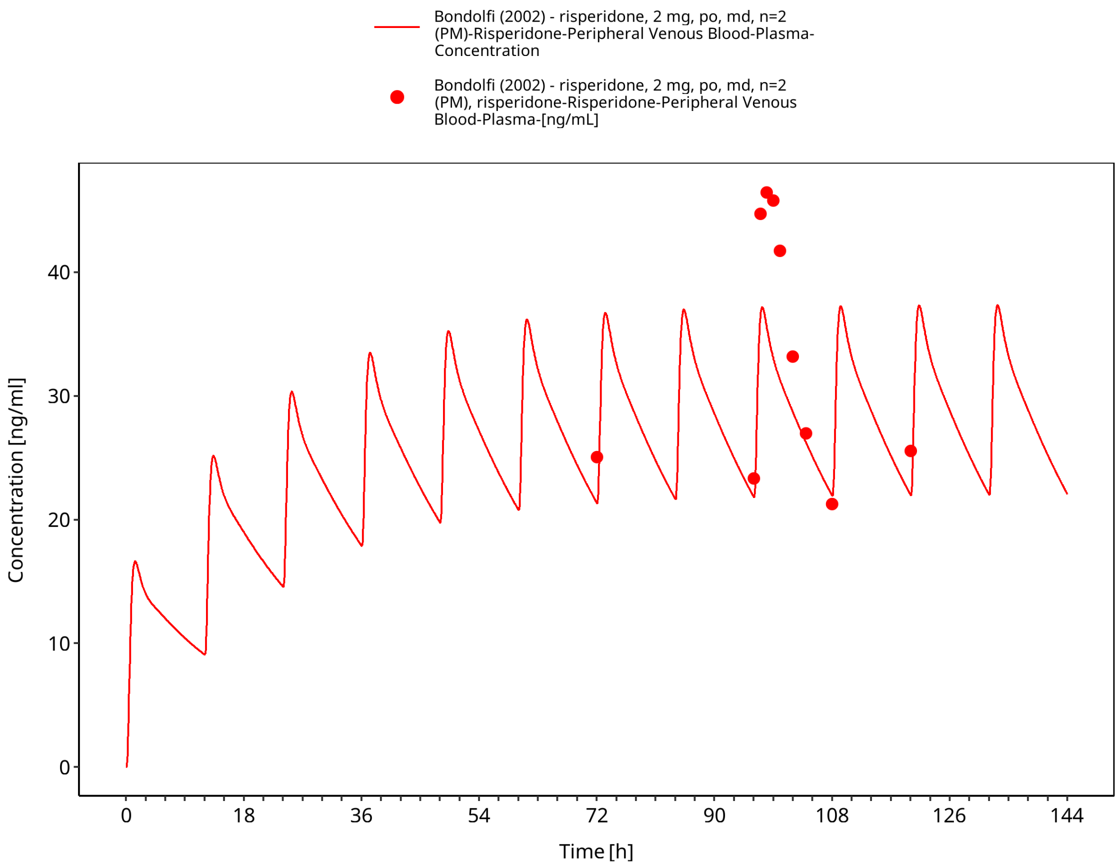

Figure 3-6: Time Profile Analysis

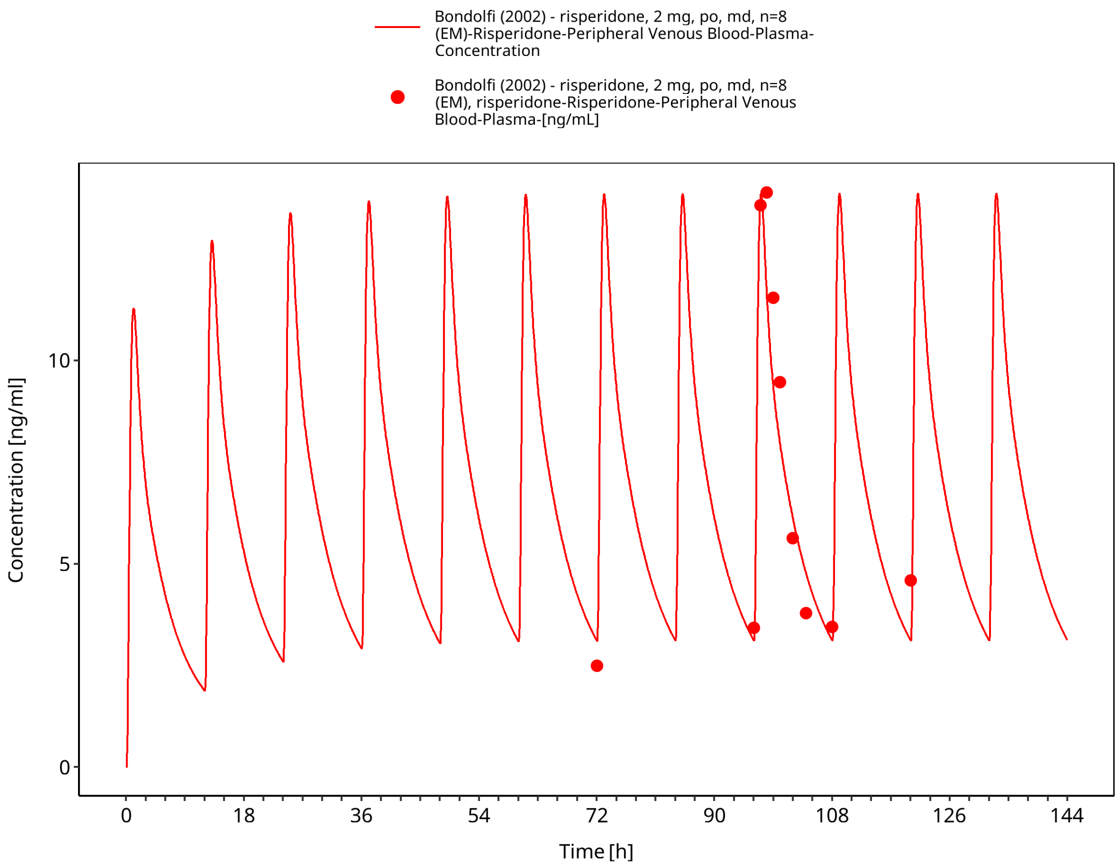

Figure 3-7: Time Profile Analysis

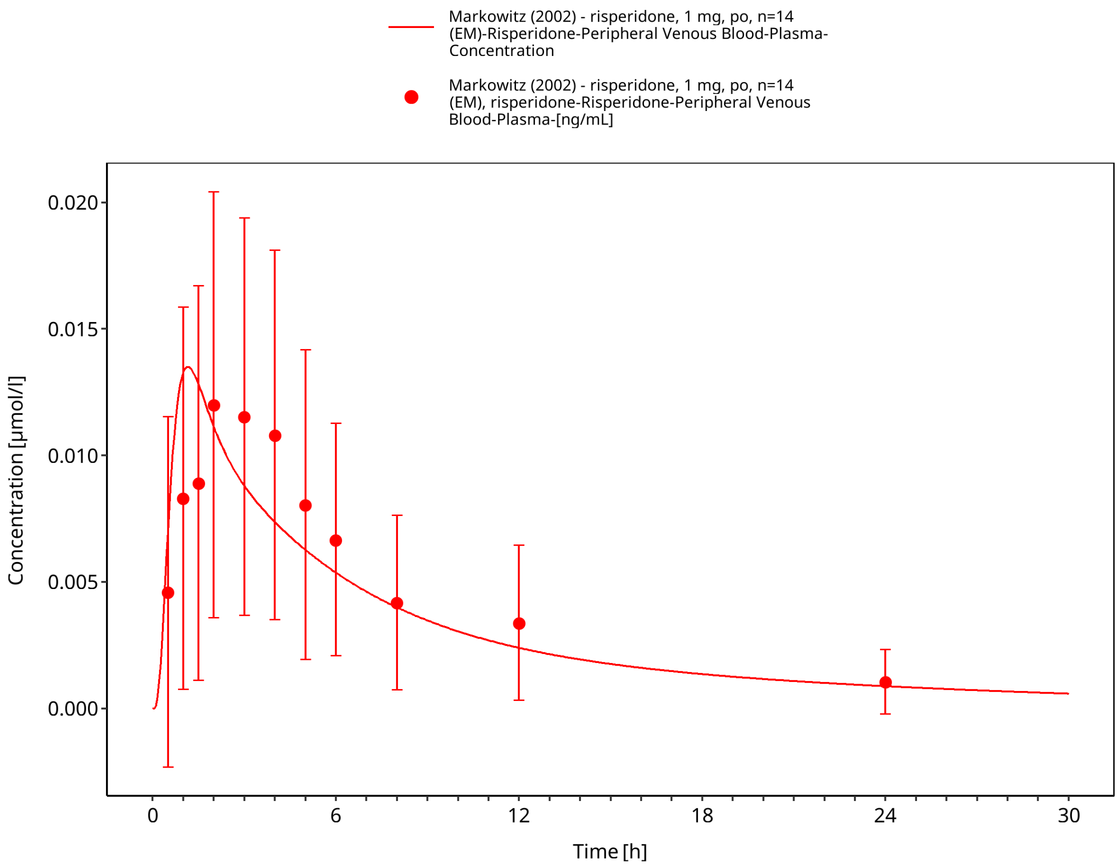

Figure 3-8: Time Profile Analysis

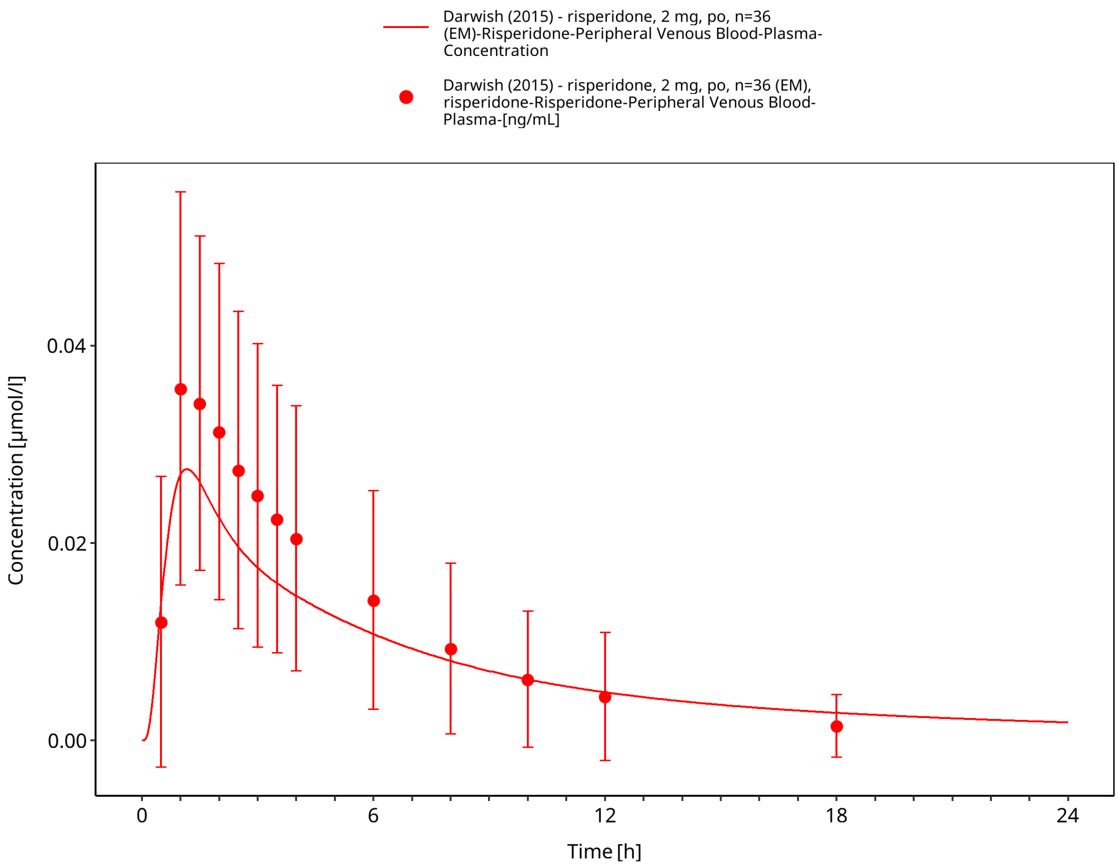

Figure 3-9: Time Profile Analysis

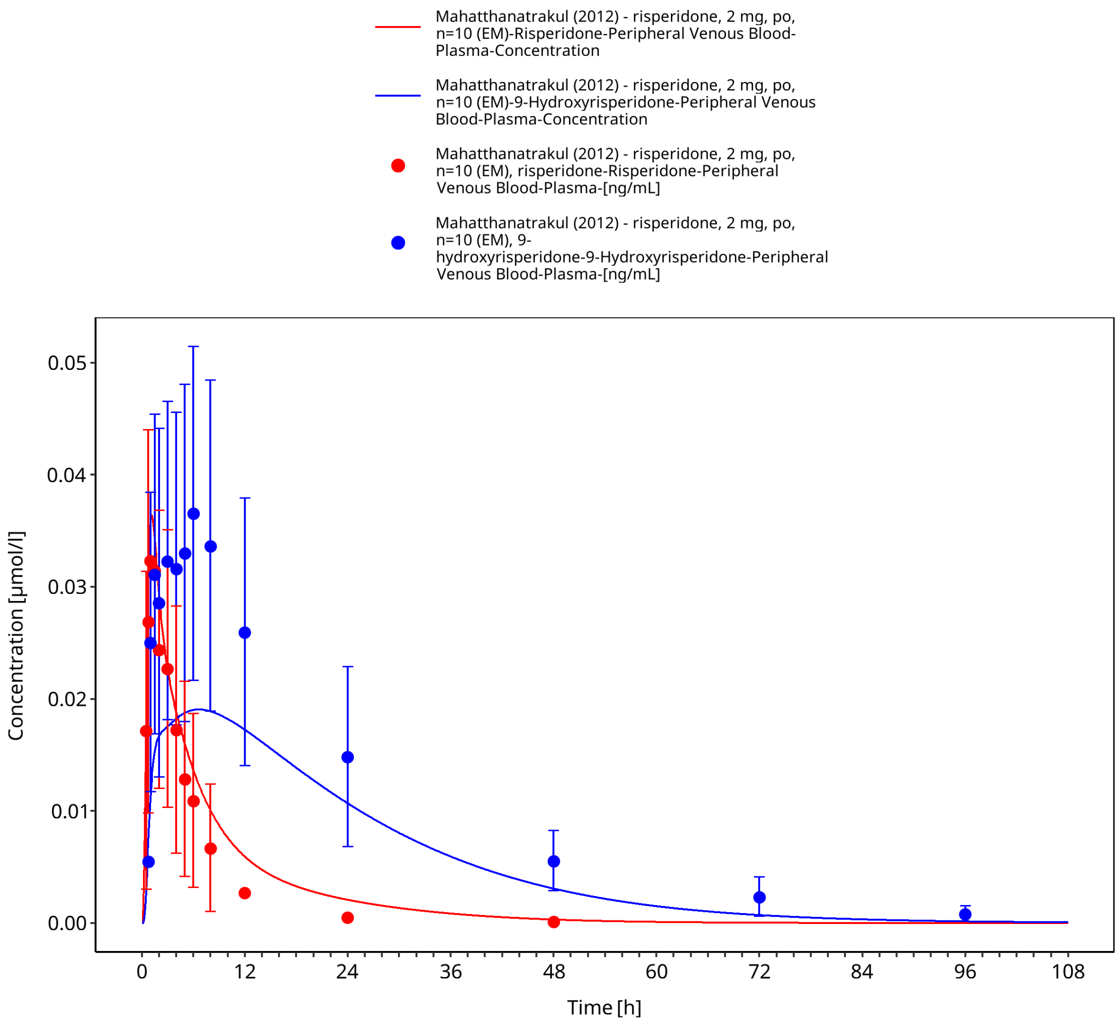

Figure 3-10: Time Profile Analysis

3.3.2 Model Verification

The following verification profiles compare the final model with independent risperidone studies across dose and CYP2D6 activity groups. Parent and 9-hydroxyrisperidone performance should be evaluated separately before considering active-moiety behavior.

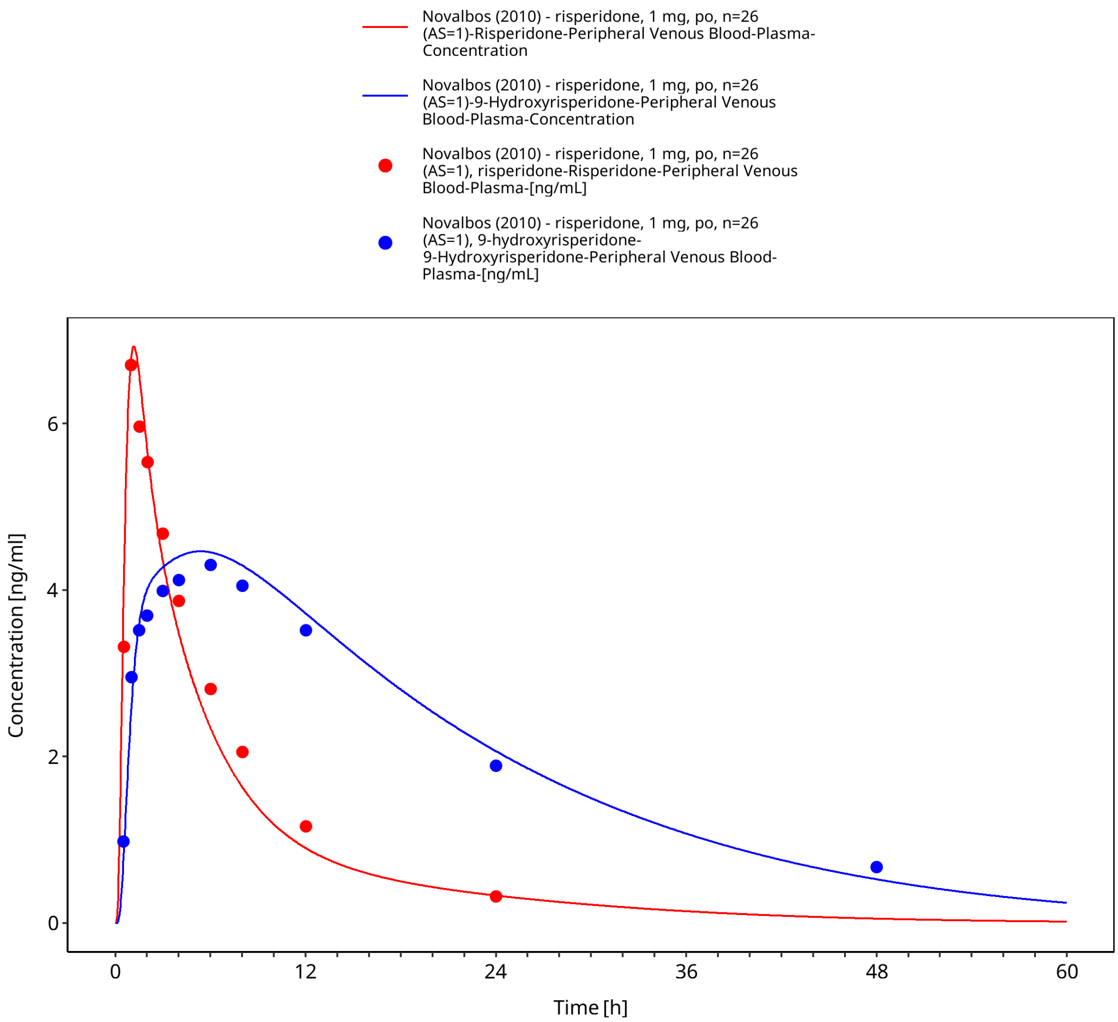

Figure 3-11: Time Profile Analysis

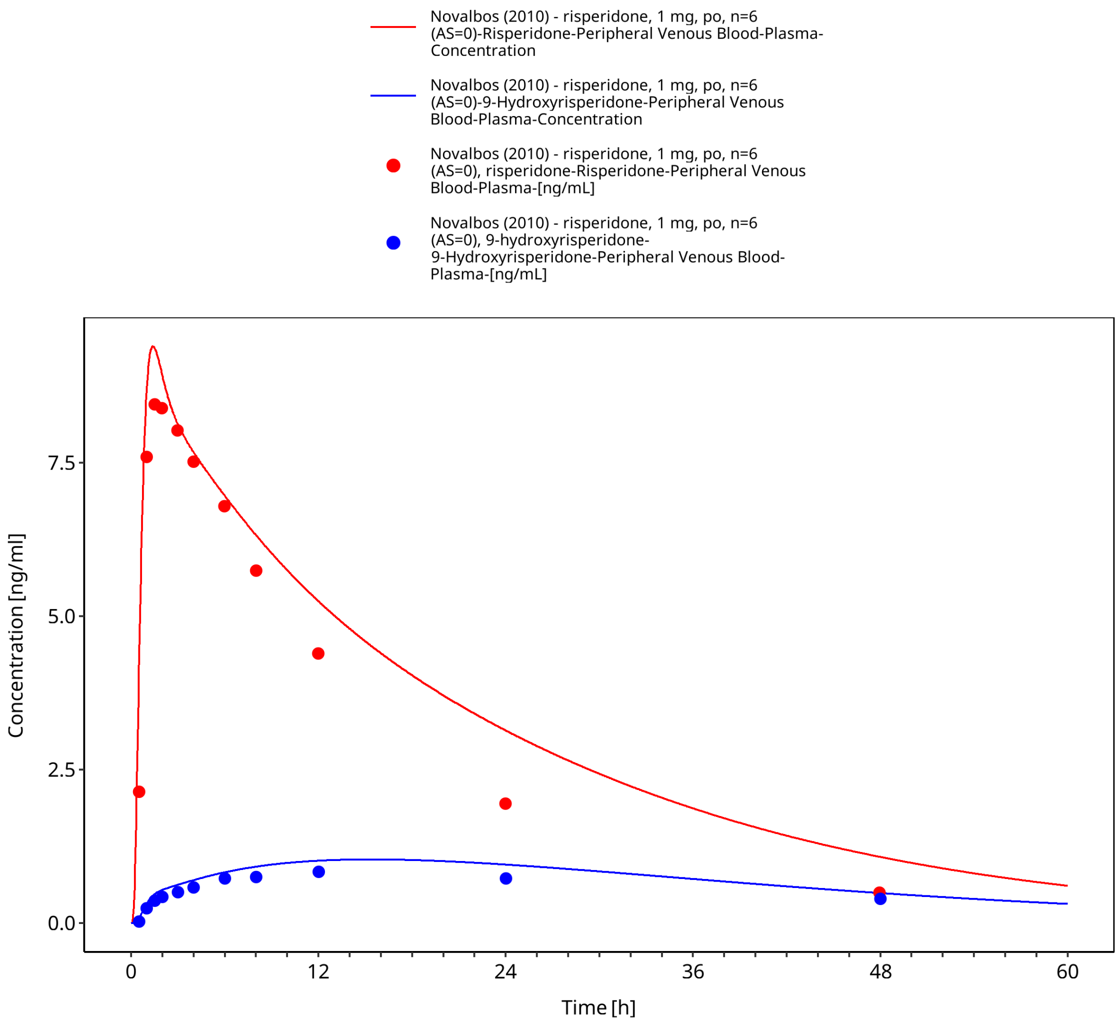

Figure 3-12: Time Profile Analysis

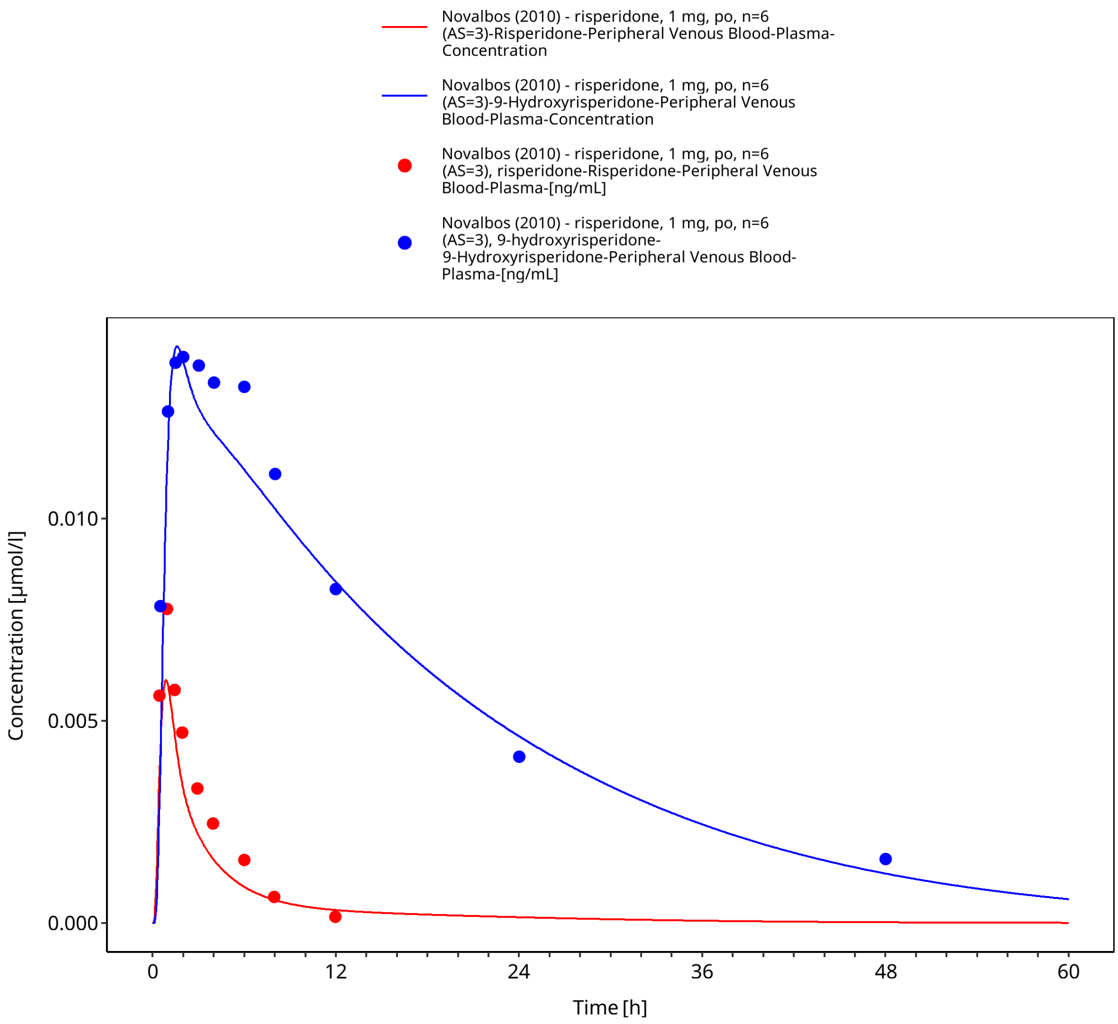

Figure 3-13: Time Profile Analysis

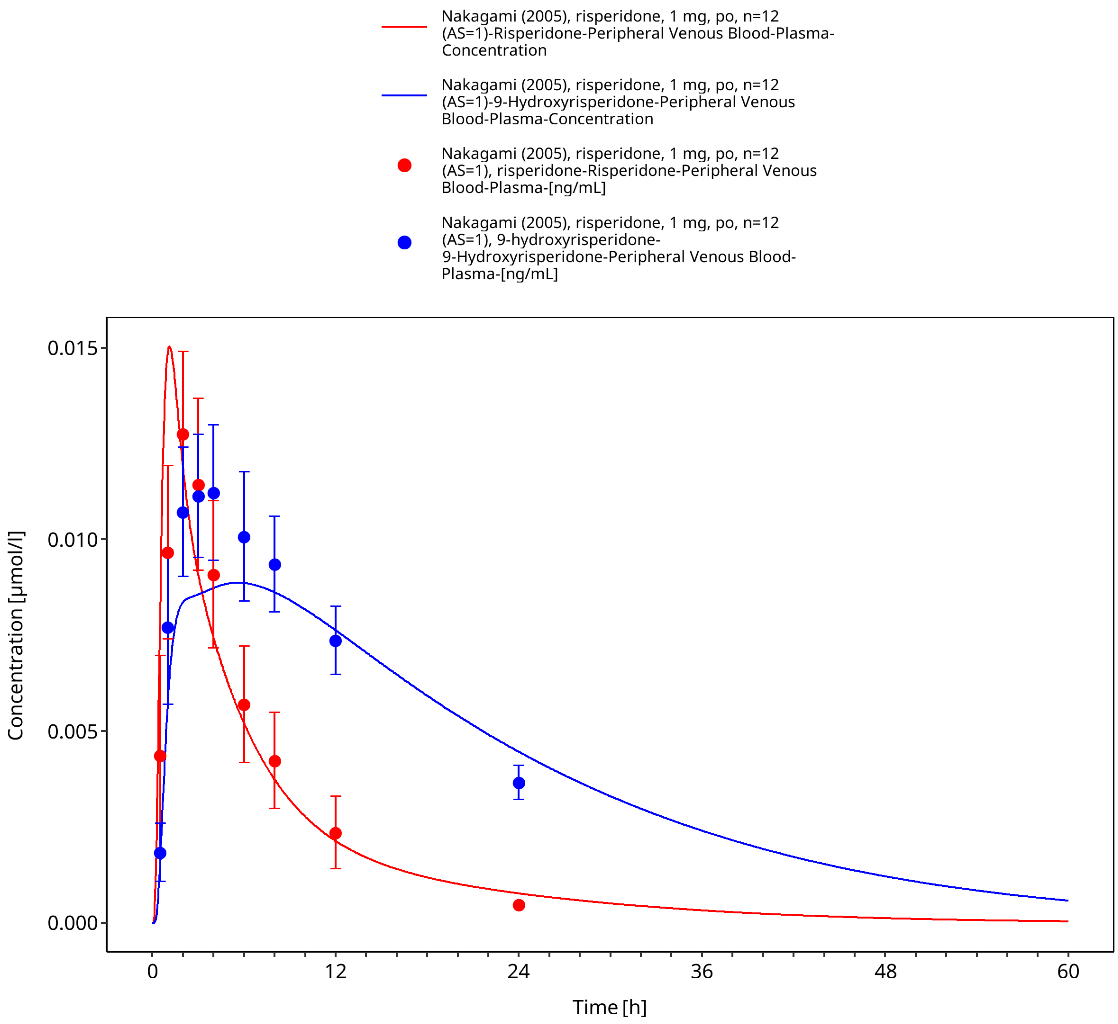

Figure 3-14: Time Profile Analysis

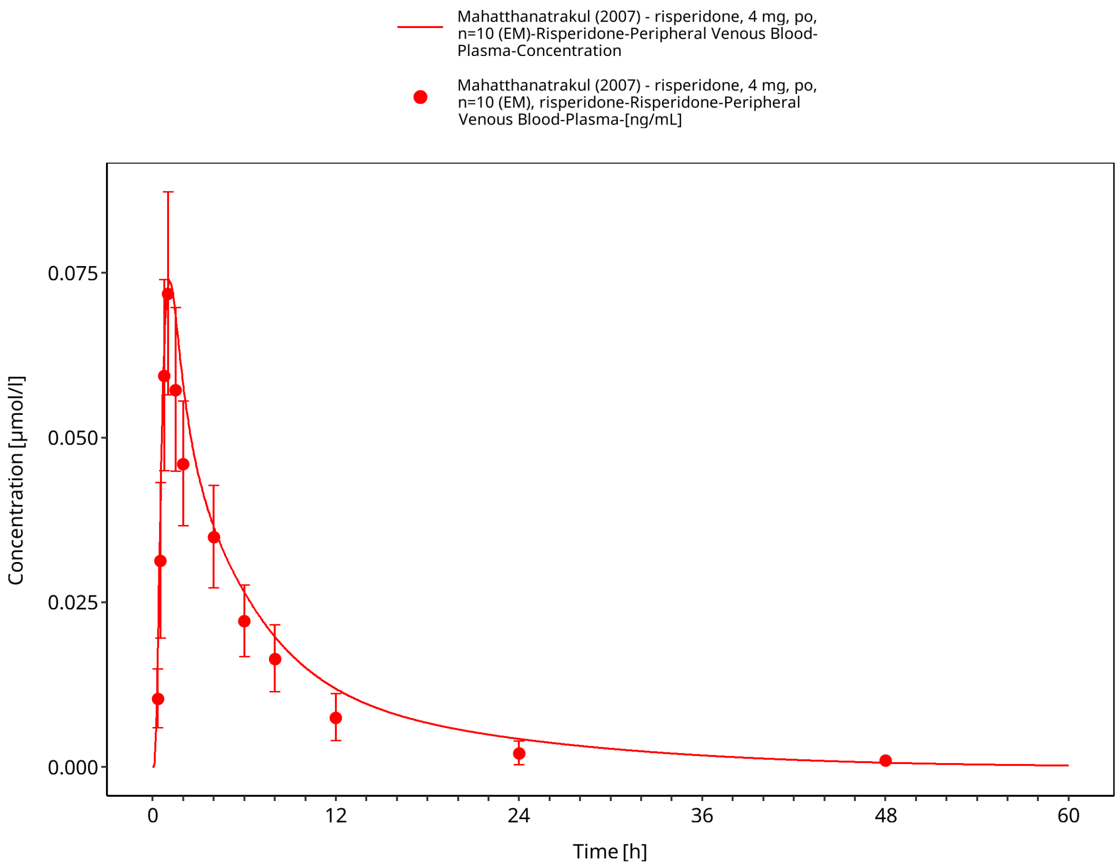

Figure 3-15: Time Profile Analysis

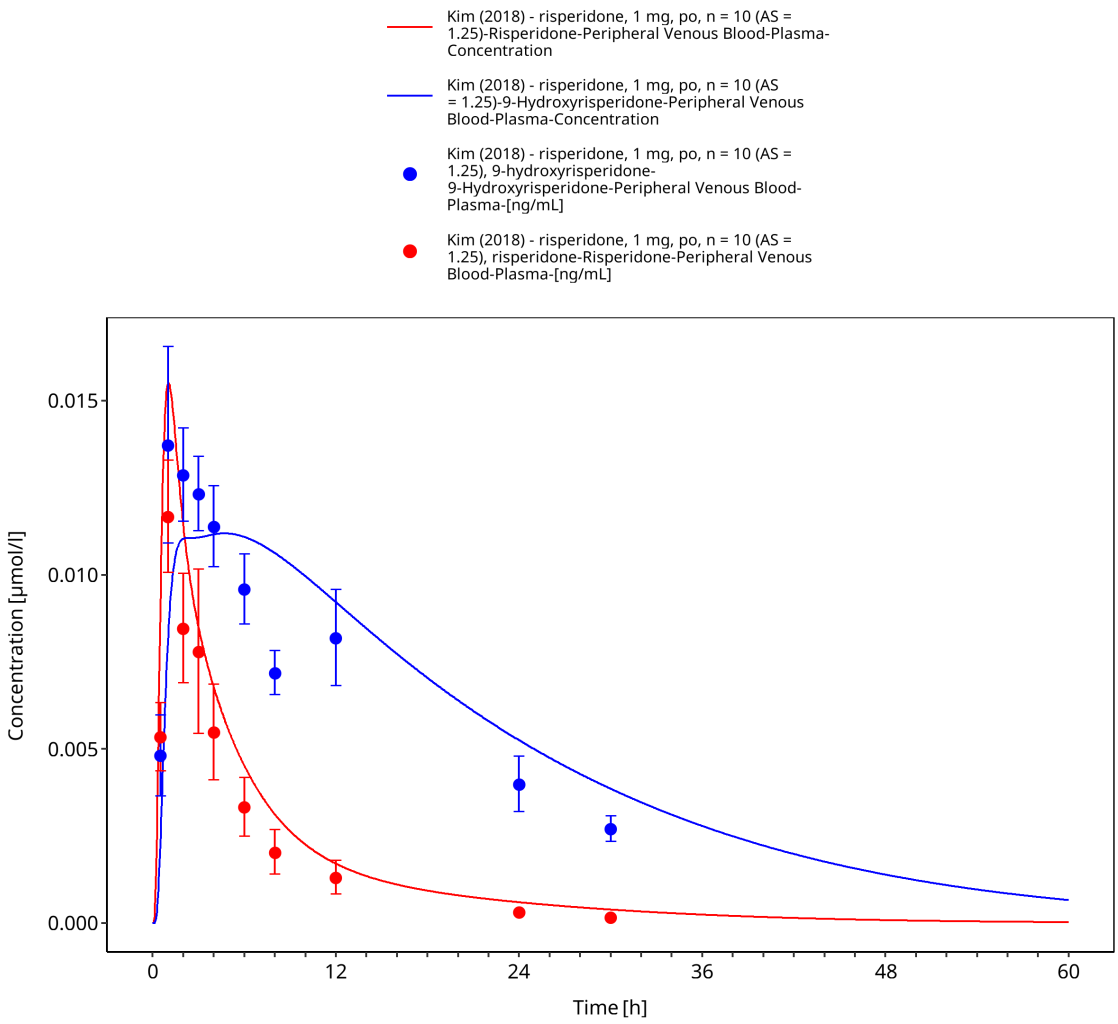

Figure 3-16: Time Profile Analysis

4 Conclusion

The presented PBPK model adequately describes the oral pharmacokinetics of risperidone and 9-hydroxyrisperidone in the evaluated adult CYP2D6 phenotype and activity-score groups.

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